# Number Sequences & Patterns

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

**Worked examples — tutor talks through these first**

**A) Easy:** Find the nth term rule for the sequence 6, 11, 16, 21, … and find the 15th term.
*Method & answer: Common difference 5, first term 6, so nth term = 5n+1. 15th term = 5×15+1 = 76.*

**B) Standard:** The nth term of a sequence is $3n^2 - 2$. Find the 4th term.
*Method & answer: $3\times4^2 - 2 = 3\times16 - 2 = 46$.*

**C) Advanced:** A sequence follows the rule 'multiply by 2 then add 5', starting at 3. Find the 4th term.
*Method & answer: 3 → 11 → 27 → 59. The 4th term is 59.*

## Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** Find the nth term rule for the sequence 4, 9, 14, 19, …

**2.** Find the 12th term of the sequence with nth term rule 4n + 3.

**3.** Which is the nth term rule for the sequence 7, 13, 19, 25…? A) 6n+1 B) 6n−1 C) 7n D) 5n+2

**4.** Find the next two terms: −12, −7, −2, __, __.

**5.** Find the missing term: 8, 17, __, 35, 44.

**6.** Select ALL the sequences below with a common difference of 9: A) 4,13,22,31 B) 5,14,24,33 C) 8,17,26,35 D) 10,19,28,38 E) 2,11,20,29

**7.** A sequence has nth term 7n − 4. Find the first three terms.

**8.** Find the next two terms: 200, 185, 170, __, __.

**9.** The nth term of a sequence is 10n − 6. Which of these is the 9th term? A) 74 B) 78 C) 84 D) 90

**10.** Find the missing terms: −5, __, 5, __, 15.

**11.** Find the next two terms: 3, 3.5, 4, 4.5, __, __.

**12.** Select ALL of these that are terms in the sequence with nth term 6n + 1: A) 13 B) 19 C) 25 D) 30 E) 37

**13.** Find the nth term rule for the sequence 100, 91, 82, 73, …

**14.** A sequence has nth term $2n^2 + 1$. Which of these is the 4th term? A) 17 B) 33 C) 9 D) 20

**15.** Find the next two terms: 2, 7, 15, 26, __, __.

## Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** The nth term of a sequence is $n^2 + 2n$. Find the first three terms.

**2.** Find the next two terms: 2, 6, 18, 54, __, __.

**6.** Select ALL the sequences below where each term is multiplied by 3 to get the next: A) 2,6,18,54 B) 5,10,15,20 C) 4,12,36,108 D) 3,9,27,81 E) 6,18,36,72

**7.** Find the next two terms: 1, 4, 10, 22, __.

**11.** A sequence has nth term $n^2 - 1$. Select ALL of these that are terms: A) 3 B) 8 C) 15 D) 20 E) 24

**12.** Find the next two terms: 800, 400, 200, __, __.

**3.** A sequence has nth term $3n^2$. Which is the 5th term? A) 15 B) 45 C) 75 D) 90

**4.** Find the missing term: 5, 15, 45, __, 405.

**5.** A sequence starts at 1000 and each term is half of the one before. Find the 5th term.

**8.** Find the nth term rule for the square numbers 1, 4, 9, 16, 25... and state the 12th term.

**9.** Which nth term rule generates the sequence 2, 8, 18, 32, 50...? A) $2n^2$ B) $n^2+1$ C) $4n-2$ D) $2n+6$

**10.** Find the missing term: 7, 21, 63, __, 567.

**13.** The nth term of a sequence is $5n^2 - 3$. Find the 3rd term.

**14.** Find the missing term: 6, 18, 54, __, 486. A) 108 B) 132 C) 150 D) 162

**15.** A geometric sequence starts at 3 and each term is multiplied by 4. Find the 5th term.

## Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** Find the nth term rule for the sequence 12, 19, 26, 33, ... and use it to find the 50th term.

**2.** A sequence has nth term $4n - 7$. For which value of n does the term first exceed 100? A) n = 26 B) n = 27 C) n = 28 D) n = 29

**3.** A theatre has 18 seats in row 1, and each row has 3 more seats than the row before, for 24 rows. How many seats are in the theatre altogether?

**4.** Select ALL of the following sequences that match the nth term rule $8n - 5$: A) 3, 11, 19, 27 B) 3, 10, 17, 24 C) 3, 11, 19, 27, 35 D) 8, 16, 24, 32 E) 3, 11, 19

**5.** A sequence's nth term is $n^2$. Find the sum of the first 6 terms of the sequence.

**6.** Find the nth term rule for the sequence 500, 485, 470, 455, ... and find the first negative term.

**7.** The nth term of a sequence is $n^2 - 5n + 6$. Which of these is the 6th term? A) 6 B) 8 C) 12 D) 16

**8.** A ball dropped from 6m bounces to 60% of its previous height each time. How high (to the nearest cm) is the 4th bounce?

**9.** Two sequences have nth term rules $6n - 1$ and $4n + 9$. For what value of n do they give the same term, and what is that term?

**10.** A sequence has nth term $3n^2 - 2n$. Select ALL of these that are terms: A) 1 B) 8 C) 20 D) 21 E) 33

**11.** A stadium's seating follows the pattern: row 1 has 40 seats, and each subsequent row has 5 fewer seats than the row before, for 6 rows. How many seats are in the stadium in total?

**12.** Find the nth term rule for the sequence 2, 9, 28, 65, ... (hint: compare to $n^3$).

**13.** A sequence has nth term $100 - 4n$. Which is the first term that is less than or equal to zero? A) the 24th term B) the 25th term C) the 26th term D) the 20th term

**14.** A savings account starts at £200 and grows by 12% each year (compound interest). What is the balance after 3 years, to the nearest pound?

**15.** A sequence of triangular numbers has nth term $n(n+1)/2$. Find the smallest n for which the triangular number exceeds 200.